

Distance Counter

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Presented here is a simple pedometer circuit. It measures the distance covered by you while walking. It may not work very well for running!

Circuit and working

Fig. 1 shows circuit diagram of the distance counter. The circuit is built around quad 2-input Schmitt trigger CD4093 (IC1), CMOS ripple carry binary counter/divider CD4024 (IC2), decade counter/divider CD4026 (IC3 and IC4), two transistors BC327 (T1,

T2) and some other components.

Gates N1 and N2 of IC1 form a monostable multivibrator that receives trigger input from tilt or mercury

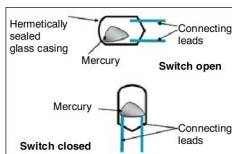


Fig. 2: Open-close operation of mercury switch

switch S1. When you lift your foot up and touch the ground back during walking, the mercury inside the switch makes a contact with its two metallic leads as shown in Fig. 2. This makes

Test Points

Test point	Details
TP0	GND, 0V
TP1	+3V when power switch S3 is closed
TP2	High when reset and remains high till 64 counts and low for the next 64 counts
TP3	Low to high when both S4 and S5 are pressed simultaneously
TP4	High when S5 is pressed to display the reading

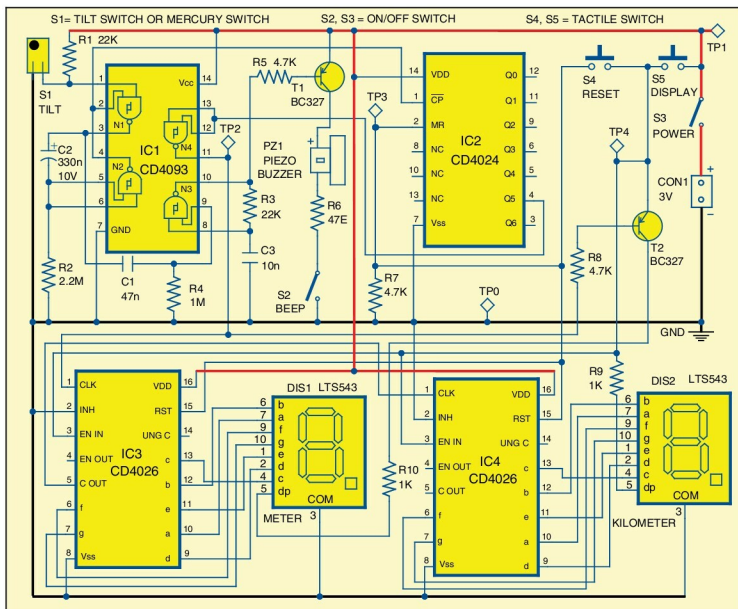


Fig. 1: Circuit diagram of the distance counter